



Fight the landlord

CS181 Final Project

钱磊 张子豪 林子修 徐景骐

Motivation

- Fight the landlord is a competitive card game with **adversarial decision-making**.
- It also has **team cooperation**, because two farmers play against one landlord.
- The game has **imperfect information**: players cannot see opponents' hands.
- The large action space makes it valuable for studying AI search and learning methods.



Settings

Problem Formulation / Environment Definition

- **Players:** 3 players: 1 landlord and 2 farmers.
- **Cards:** 54 cards, including two Jokers.
- **Goal:** Be the first side to play all cards.
- **State:** Player hand, public play history, current turn, and remaining card counts.
- **Action:** Any legal card combination that can be played.
- **Number of states:** Approximately 10^{83} possible states.



GameRules

Basic Rules

- **Card order:** Big Joker > Small Joker > 2 > A > K > Q > J > 10 > 9 > 8 > 7 > 6 > 5 > 4 > 3.
- Suits do not affect card strength.
- Each player receives **17 cards**.
- The remaining **3 bottom cards** are given to the landlord after bidding.
- The landlord plays alone with **20 cards**.
- The two farmers cooperate as one side.

GameRules

Advanced Combinations

- **Single:** one card, such as 7.
- **Pair:** two cards of the same rank, such as 8-8.
- **Triple:** three cards of the same rank, such as 5-5-5.
- **Triple with one:** three cards plus one extra card, such as 5-5-5 + 9.
- **Triple with pair:** three cards plus one pair, such as 5-5-5 + 9-9.
- **Straight:** consecutive single cards, such as 3-4-5-6-7.
- **Bomb:** four cards of the same rank.
- **Rocket:** Big Joker + Small Joker.

GameRules

Scores

- At the beginning, players bid **1, 2, or 3 points**.
- The highest bidder becomes the landlord.
- The bid is the base score.
- Each bomb doubles the score: $\times 2$.
- A rocket also doubles the score: $\times 2$.
- Final score = base score $\times$ all multipliers.

Implement methods

1. Adversarial Search

- **Minimax Algorithm**
 - Model the landlord and farmers as two opposing sides.
- **Depth-limited Adversarial Search**
 - Limit search depth to reduce computation.
- **Heuristic State Evaluation**
 - Evaluate hand strength, remaining cards, and possible control.
- **Expectimax Algorithm**
 - Consider uncertainty from hidden opponent cards.

Implement methods

2. CSP

- Use Constraint Satisfaction Problems to model legal card grouping.
- Find possible partitions of a hand into legal combinations.
- Prefer fewer groups and stronger combinations.
- Help the agent choose efficient card structures.

Implement methods

3. Inference

- **Exact Inference**
 - Track all played cards exactly.
 - Remove known cards from the unknown card pool.
- **Approximate Inference**
 - Estimate possible opponent hands.
 - Use sampling when exact calculation is too expensive.
 - Update beliefs from bidding and playing behavior.

Implement methods

4. MDP

- **Markov Decision Process (MDP)**
 - Define states, actions, transitions, and rewards.
- **State Value Function**
 - Estimate how good a game state is.
 - Use value estimates to guide move selection.

Implement methods

5. Reinforcement Learning

- **Q-Learning**
 - Learn action values through self-play.
- **ϵ -Greedy Policy**
 - Balance exploration and exploitation.
- **Policy Search**
 - Directly optimize the agent's playing strategy.



Thank you